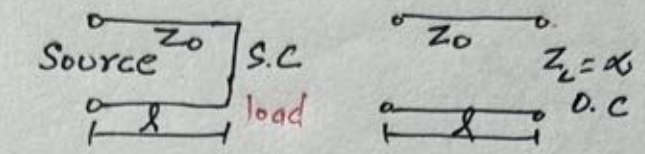
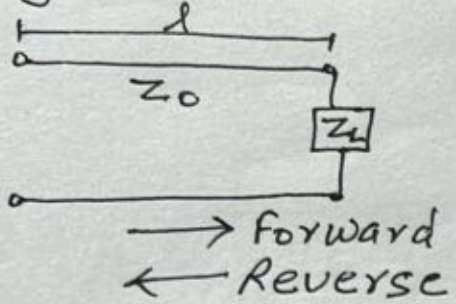
Stub or stub matching

* It is a short circuit or open circuit Transmission line. E.g.



* Why do we use stub ?



if $Z_L \neq Z_0$

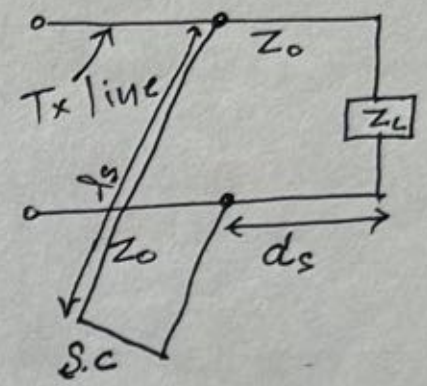
Then Reflection occurs in Transmission line

* If $\Gamma > 0$, max Power will not transfer to load

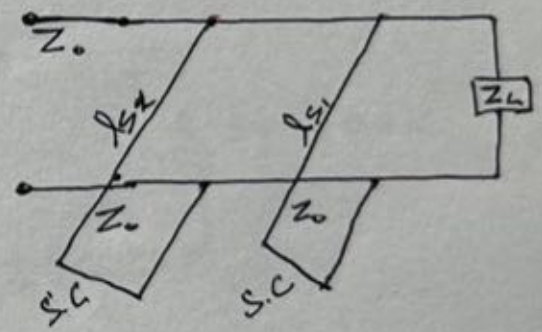
* To match Z_L and Z_0, we use stub in transmission line, so that max Power of source will be transferred to load end.

Types of stub

i) Single stub



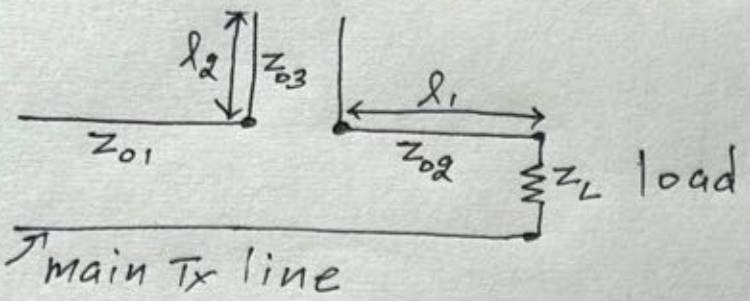
ii) Double stub



* Connection of stub: i) Series connection → Z
  ii) Parallel connection → Y

* Short circuit stub is better than open circuit stub.

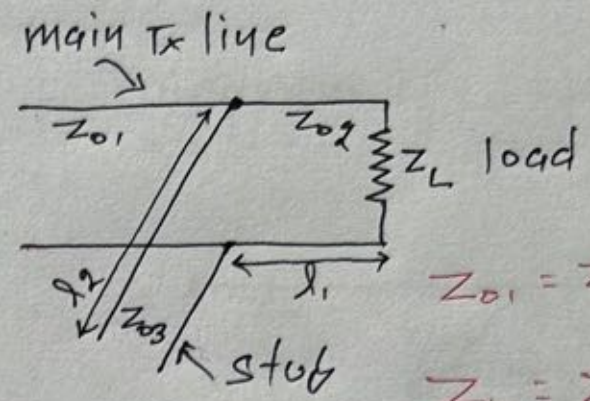
open circuit series stub



$Z_{01} = Z_{02} = Z_{03} = Z_0$

$Z_L = Z_0$

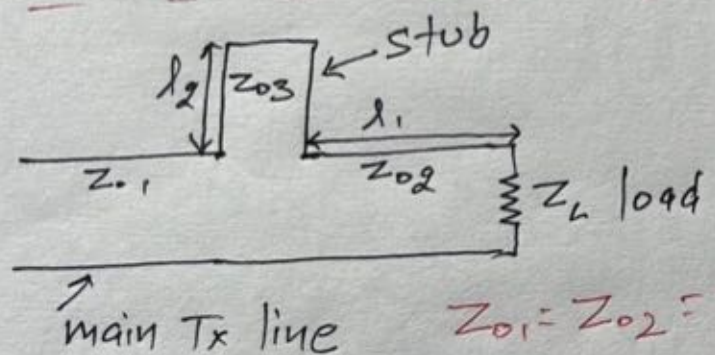
open circuit parallel stub



$Z_{01} = Z_{02} = Z_{03} = Z_0$

$Z_L = Z_0$

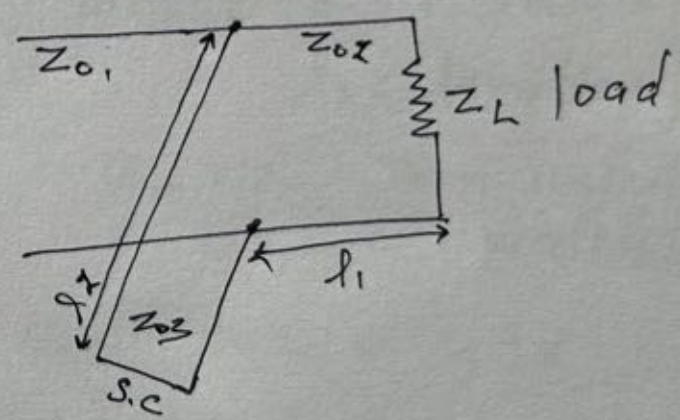
short circuit series stub



$Z_{01} = Z_{02} = Z_{03} = Z_0$

$Z_L = Z_0$ ✓

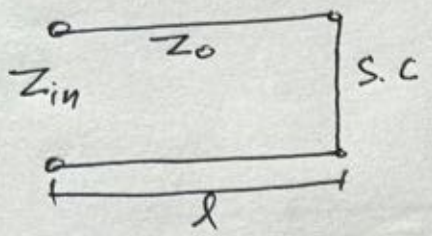
short circuit parallel stub



Concept of Impedance matching using Single stub

* We know

(I)



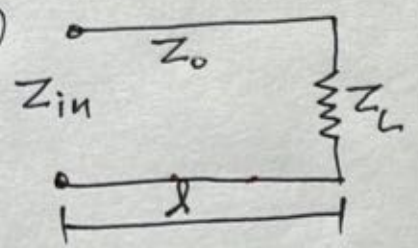
$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l} \right]$$

$Z_L = 0$

$$Z_{in} = jZ_0 \tan\beta l$$

* For S.C line, Z_{in} is allways reactance

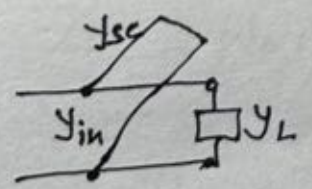
(II)



$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l} \right]$$

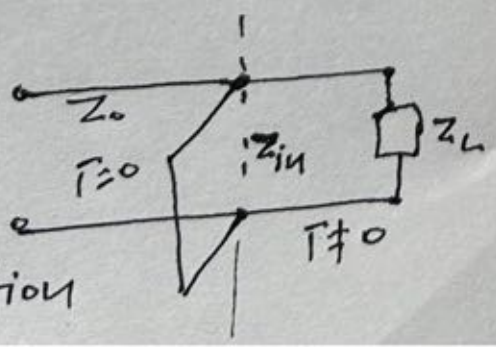
$$Z_{in} = f(l)$$

(III) Admittance of Parallel connection



$Y_{in} =$

(IV) if

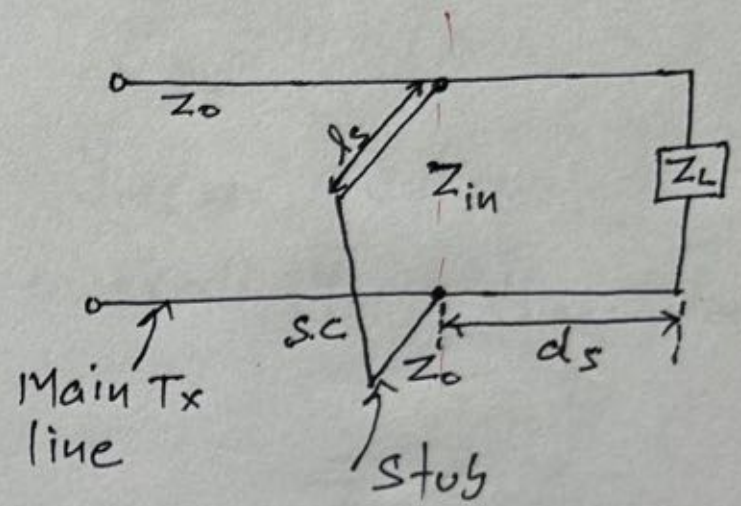


$Z_L = Z_{in}$

Then $\Gamma = 0$

No reflection

Diagram of single stub



where d_s = Position of stub

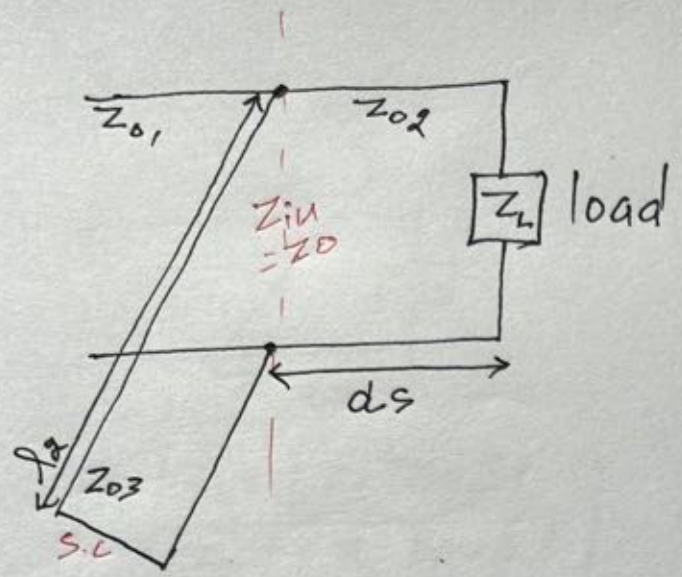
l_s = length of stub

$Z_{in} = Z_0$. Then impedance will match.

Y_{in} due to $Z_L \Rightarrow 1 + jx$

Y_{in} due to stub $\Rightarrow -jx$

Net $Y_{in} = 1 + jx - jx = 1$



* Term used

$\bar{Z}_L$ = Normalized load impedance

$$\bar{Z}_L = \frac{Z_L}{Z_0}$$

Z_{in} = input impedance

$\bar{Z}_{in}$ = Normalized input impedance

$$\bar{Z}_{in} = \frac{Z_{in}}{Z_0} = 1$$

if Parallel connection

* study admittance $y_{in(L)}$ or $\bar{y}_{in(L)}$

$\bar{y}_{in}$ at ds distance from Z_L

$$\bar{y}_{in(L)} = 1 + jx$$

$$\bar{y}_{in(S)} = -jx$$

overall $\bar{y}_{in} = \bar{y}_{in(L)} + \bar{y}_{in(S)}$

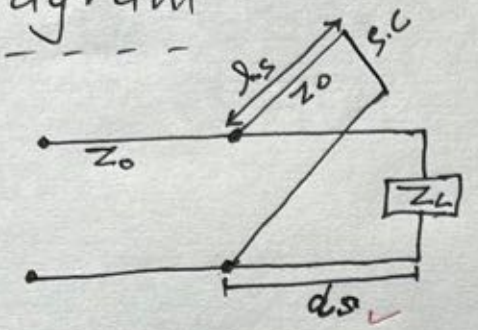
$= 1$

$\bar{Z}_{in} = 1$

$$\boxed{Z_{in} = Z_0}$$

Derivation of single stub matching

Diagram



$$Z_{in} = Z_0\left[\frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}\right]$$

$$\frac{Z_{in}}{Z_0} = \left[\frac{\frac{Z_L}{Z_0} + j\tan\beta l}{1 + j\frac{Z_L}{Z_0}\tan\beta l}\right]$$

$$\bar{Z}_{in} = \frac{\bar{Z}_L + j\tan\beta l}{1 + j\bar{Z}_L \tan\beta l}$$

$$\frac{1}{\bar{y}_{in}} = \frac{\frac{1}{\bar{y}_L} + j\tan\beta l}{1 + j\frac{1}{\bar{y}_L}\tan\beta l}$$

$$\frac{1}{\bar{y}_{in}} = \frac{1 + j\bar{y}_L \tan\beta l}{\bar{y}_L + j\tan\beta l}$$

$$\bar{y}_{in} = \frac{\bar{y}_L + j\tan\beta l}{1 + j\bar{y}_L \tan\beta l}$$

$$\bar{y}_{in} = \frac{\bar{y}_L + j\tan\beta l}{1 + j\bar{y}_L \tan\beta l} \times \frac{1 - j\bar{y}_L \tan\beta l}{1 - j\bar{y}_L \tan\beta l}$$

$$\bar{y}_{in} = \frac{\left(\bar{y}_L + \bar{y}_L \tan^2\beta l\right) + \left(\tan\beta l - \bar{y}_L^2 \tan\beta l\right)}{1 + \bar{y}_L^2 \tan^2\beta l}$$

Real part of $\bar{y}_{in} = \dfrac{\bar{y}_L + \bar{y}_L \tan^2\beta l}{1 + \bar{y}_L^2 \tan^2\beta l}$ must be 1

$$\frac{\bar{y}_L + \bar{y}_L \tan^2\beta l}{1 + \bar{y}_L^2 \tan^2\beta l} = 1$$

$$\bar{y}_L + \bar{y}_L \tan^2\beta l = 1 + \bar{y}_L^2 \tan^2\beta l$$

$$\bar{y}_L \tan^2\beta l - \bar{y}_L^2 \tan^2\beta l = 1 - \bar{y}_L$$

$$\bar{y}_L \tan^2\beta l\left(\cancel{1 - \bar{y}_L}\right) = \left(\cancel{1 - \bar{y}_L}\right)$$

$$\bar{y}_L \tan^2\beta l = 1$$

$$\tan^2\beta l = \bar{Z}_L = \frac{Z_L}{Z_0} \Rightarrow \tan\beta l = \sqrt{\frac{Z_L}{Z_0}}$$

$$\beta l = \tan^{-1}\left(\sqrt{\frac{Z_L}{Z_0}}\right)$$

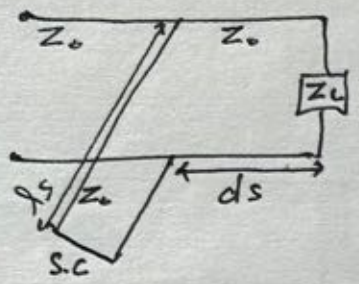
$$\frac{2\pi}{\lambda} \times l = \tan^{-1}\left(\sqrt{\frac{Z_L}{Z_0}}\right)$$

$$\boxed{l = \frac{\lambda}{2\pi}\tan^{-1}\left(\sqrt{\frac{Z_L}{Z_0}}\right)}$$

Position of stub

Derivation of single stub matching

$$\bar{y}_{in} = \frac{\left(\bar{y}_L + \bar{y}_L \tan^2\beta l\right)}{1+\bar{y}_L^2 \tan^2\beta l} + j\frac{\left(\tan\beta l - \bar{y}_L^2 \tan\beta l\right)}{1+\bar{y}_L^2 \tan^2\beta l}$$



$$\tan\beta l = \sqrt{\frac{Z_L}{Z_0}}$$

$$\bar{y}_L = \frac{1}{\bar{z}_L} = \frac{Z_0}{Z_L}$$

$$Z_{in} \text{ due to stub} = Z_0\left[\frac{Z_L + jZ_0\tan\beta l}{Z_0 + jZ_L\tan\beta l}\right]$$

$$Z_L = 0$$

$$Z_{in(s)} = jZ_0\tan\beta l$$

$$\frac{Z_{in(s)}}{Z_0} = j\tan\beta l$$

$$\bar{Z}_{in(s)} = j\tan\beta l$$

$$\boxed{\bar{y}_{in(s)} = -j\cot\beta l}$$

$$\cot\beta l = \frac{\tan\beta l - \bar{y}_L^2\tan\beta l}{1+\bar{y}_L^2\tan^2\beta l}$$

$$\cot\beta l = \frac{\tan\beta l\left(1-\bar{y}_L^2\right)}{1+\bar{y}_L^2\tan^2\beta l}$$

$$\cot\beta l = \frac{\sqrt{\frac{Z_L}{Z_0}} \times \left(1-\frac{Z_0^2}{Z_L^2}\right)}{1+\frac{Z_0^2}{Z_L^2}\times\frac{Z_L}{Z_0}}$$

$$\cot\beta l = \frac{\sqrt{\frac{Z_L}{Z_0}} \times \left(\frac{Z_L^2-Z_0^2}{Z_L^2}\right)}{1+\frac{Z_0}{Z_L}}$$

$$\cot\beta l = \frac{\sqrt{\frac{Z_L}{Z_0}} \times \left(\frac{(Z_L-Z_0)\cancel{(Z_L+Z_0)}}{Z_L^2}\right)\times \cancel{Z_L}}{\cancel{(Z_L+Z_0)}}$$

$$\cot\beta l = \sqrt{\frac{Z_L}{Z_0}} \times \left(\frac{Z_L-Z_0}{Z_L}\right)$$

$$\cot\beta l = \sqrt{\frac{\cancel{Z_L}}{Z_0\times Z_L^{\cancel{2}}}} \times Z_L - Z_0$$

$$\cot\beta l = \frac{Z_L-Z_0}{\sqrt{Z_L Z_0}}$$

$$\tan\beta l = \frac{\sqrt{Z_L Z_0}}{Z_L-Z_0}$$

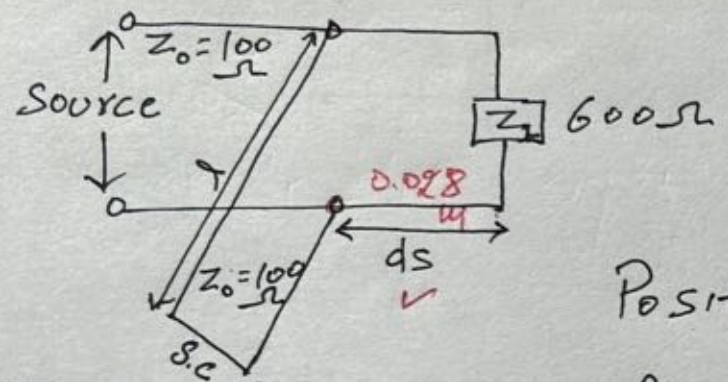
$$\beta l = \tan^{-1}\left(\frac{\sqrt{Z_L Z_0}}{Z_L-Z_0}\right)$$

$$\boxed{l_s = \frac{\lambda}{2\pi}\tan^{-1}\left(\frac{\sqrt{Z_L Z_0}}{Z_L-Z_0}\right)}$$

Stub length

Problem on single stub matching

Q.1 → 100 Ω line with air as dielectric medium is terminated by load of 600 Ω. Source is excited by 2 GHz at generator. The the find i) Position of stub
ii) length of stub of 100 Ω impedance



$Z_L = 600\ \Omega$

$Z_0 = 100\ \Omega$

$f = 2 \times 10^9$ Hz

$f = 2 \times 10^9$ Hz

$c = 3 \times 10^8$ m/s

$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{2 \times 10^9}$

$\lambda = 0.15$ meter

$180^\circ \rightarrow \pi$

$1 \rightarrow \pi/180$

$67.79^\circ \rightarrow \frac{\pi}{180} \times 67.79^\circ$

Position of stub $= \frac{\lambda}{2\pi} \tan^{-1}\left(\sqrt{\frac{Z_L}{Z_0}}\right)$

Position $= \frac{0.15}{2\pi} \tan^{-1}\left(\sqrt{\frac{600}{100}}\right)$

$= \frac{0.15}{2\pi} \times 67.79^\circ \times \frac{\pi}{180}$

Position $= \frac{0.15}{2\pi} \times 67.79 \times \frac{\pi}{180}$

Position of stub $= \boxed{0.028 \text{ meter}}$

$= 2.8$ cm

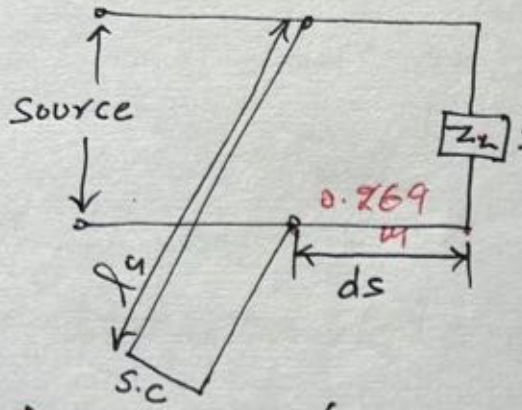
$l_s = \frac{\lambda}{2\pi} \tan^{-1}\left(\frac{\sqrt{Z_L Z_0}}{Z_L - Z_0}\right)$

$= \frac{0.15}{2\pi} \times 26.1^\circ \times \frac{\pi}{180^\circ}$

$\boxed{l_s = 0.011 \text{ meter}}$

length of stub

Q.2 A dipole Antenna having input impedance of 200 Ω is to be matched at freq of 100 MHz to Tx line of characteristic impedance of 500 Ω with help of short circuit stub. Determine the location & length of the stub



$Z_0 = 500\,\Omega$

$Z_L = 200\,\Omega$

$f = 100 \times 10^6$ Hz

$f = 10^8$ Hz

$\lambda = \frac{c}{f} = \frac{3 \times \cancel{10^8}}{\cancel{10^8}}$

$\lambda = 3$ meter

$$\text{location} = \frac{\lambda}{2\pi} \tan^{-1}\left(\sqrt{\frac{Z_L}{Z_0}}\right)$$

$$= \frac{3}{2\pi} \tan^{-1}\left(\sqrt{\frac{200}{500}}\right)$$

$$= \frac{3}{\cancel{2\pi}} \times 32.31^\circ \times \frac{\cancel{\pi}}{180}$$

$$\boxed{\text{location} = 0.269 \text{ meter}}$$

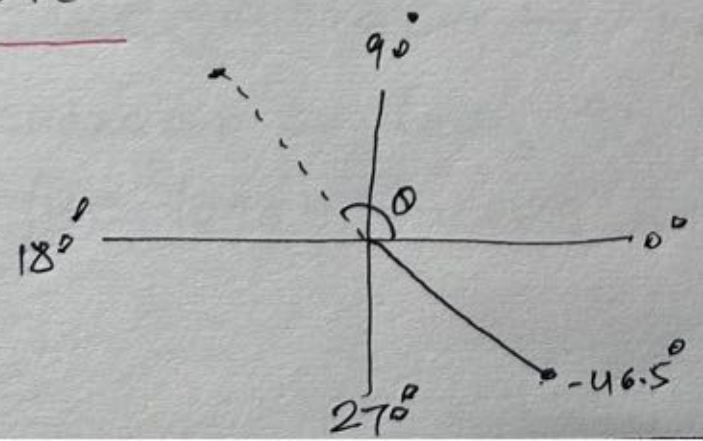
$$l_s = \frac{\lambda}{2\pi} \tan^{-1}\left(\frac{\sqrt{Z_L Z_0}}{Z_L - Z_0}\right)$$

$$= \frac{3}{2\pi} \tan^{-1}\left(\frac{316.23}{-300}\right)$$

$$= \frac{3}{2\pi} \times (-46.5)^\circ$$

$$= \frac{3}{\cancel{2\pi}} \times 133.5^\circ \times \frac{\cancel{\pi}}{180}$$

$$\boxed{l_s = 1.1125 \text{ meter}}$$



$180 - 46.5^\circ$

$= 133.5^\circ$